SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: CRYPTOGRAPHY AND NETWORK SECURITY
COURSE CODE: CSA5116

COURSE OUTCOME COVERED

CO2: Analyze Public Key crypto system strategies using RSA and key Exchange for Encryption algorithm to strengthen information security. (BL4,BL5)

Assessment Tool Weightage : 20%

QUESTIONS: **Total Marks: 50**

Annexure A

Scenario-Based

Code of Conduct:

*I **M.SADHANA[192524386]**(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Annexure A
Scenario-based

1. Secure Data Transmission using Block Cipher
A company is transmitting repeated financial data blocks over a network. Initially, they used ECB mode and noticed patterns in ciphertext.

A. Explain why this issue occurs.
B. Suggest a more secure block cipher mode and justify your choice based on the scenario.

2. Choosing an Encryption Algorithm for Banking System
A banking application needs high security and moderate performance for encrypting customer transactions. The available options are DES, 3-DES, and Blowfish.

A. Identify the most suitable algorithm for this scenario.
B. Justify your choice based on security strength and efficiency.

3. Key Exchange in an Unsecured Network
Two users want to establish a secure communication channel over a public network using Diffie-Hellman key exchange. However, the network is vulnerable to attacks.

A. Explain how the key exchange works in this scenario.
B. Identify a possible security threat and suggest a solution to mitigate it.

4. Public Key Cryptography for Email Security
An organization plans to implement RSA for secure email communication.

A. Describe how RSA can be used for encryption and decryption in this scenario.
B. Explain how key distribution and management play a role in maintaining security.

5. Cryptography for IoT Devices
A smart home system uses low-power IoT devices that need secure communication. The designer must choose between RSA and Elliptic Curve Cryptography (ECC).

A. Analyze the constraints of IoT devices.
B. Recommend a suitable cryptographic approach and justify your answer.

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

\\

ASSESSMENT OVERVIEW

Learning focus

- Confidentiality: preventing unauthorized people from reading protected information.
- Integrity: detecting unauthorized changes to information.
- Authentication: verifying the identity of a communicating user or system.
- Key management: protecting, distributing, rotating, and revoking cryptographic keys.

QUESTION 2

Choosing an Encryption Algorithm for a Banking System

Scenario: A banking application needs high security and moderate performance for encrypting customer transactions. The available options are DES, 3-DES, and Blowfish.

A. Identify the most suitable algorithm for this scenario.

Among the three given algorithms, **Blowfish** is the most suitable choice if we must select only from the listed options. It supports variable key lengths up to 448 bits and can provide good software performance. Its longer key options offer better protection than DES, which uses only a 56-bit key. However, Blowfish is not recommended for new banking systems today. A modern algorithm such as AES-GCM is preferred for real banking transactions.

C.Justify your choice based on security strength and efficiency.

1. Data Encrption Standard (DES)

DES is a symmetric-key encryption algorithm that uses an effective key length of 56 bits. Modern computers can break DES through brute-force attacks by trying possible keys. Therefore, DES does not provide sufficient security for confidential banking information and should not be used in modern banking systems.

2. Triple DES (3-DES)

Triple DES applies the DES operation three times to improve security compared with single DES. However, performing encryption multiple times makes it slower and less efficient than modern algorithms. It also uses a 64-bit block size, which creates limitations when encrypting large amounts of data. Therefore, 3-DES is considered a legacy algorithm and is unsuitable for new high-volume banking applications.

3. Blowfish

Blowfish is a symmetric block cipher that supports variable key lengths from 32 bits up to 448 bits. Its flexible key length and efficient software implementation make it more attractive than DES for the given choices. However, Blowfish has a 64-bit block size. When large volumes of financial data are encrypted, this limitation can increase the risk of repeated ciphertext blocks. Hence, Blowfish is not ideal for modern high-volume banking applications.

Comparison of Algorithms

Algorithm	Security	Efficiency	Suitability
DES	Weak 56-bit key	Fast but insecure	Not suitable
3-DES	Stronger than DES but outdated	Slower	Avoid for new systems
Blowfish	Variable key length; 64-bit block limitation	Good software performance in some uses	Best among the given options only

Recommended Approach for Banking

For a new banking application, **AES-GCM** is recommended instead of DES, 3-DES, or Blowfish. AES-GCM provides authenticated encryption, which protects the confidentiality of customer transactions and helps detect unauthorized changes.

The bank should also generate strong cryptographic keys, store them securely, restrict access to authorized users, rotate keys according to policy, and never reuse the same AES-GCM nonce with the same key.

Conclusion

Among DES, 3-DES, and Blowfish, Blowfish is the most reasonable choice when restricted to the given options. DES has an inadequate key length, while 3-DES is slower and outdated for new applications. However, Blowfish's 64-bit block size limits its suitability for high-volume financial data. Therefore, a modern banking system should use AES-GCM along with secure key management, access control, and regular security reviews.

References

- NIST, FIPS 197: Advanced Encryption Standard (AES).
- NIST, SP 800-38A: Recommendation for Block Cipher Modes of Operation.
- NIST, SP 800-57: Recommendation for Key Management

QUESTION 3

Key Exchange in an Unsecured Network

Scenario: Two users want to establish a secure communication channel over a public network using Diffie–Hellman key exchange. However, the network is vulnerable to attacks.

A. Explain how the key exchange works in this scenario.

Diffie–Hellman (DH) is a cryptographic key exchange technique that allows two users to establish a common secret key over a public network without directly transmitting the secret key. It is widely used as part of secure communication protocols.

Steps involved in Diffie–Hellman Key Exchange:

Step 1: Agree on public parameters

Alice and Bob agree on two public values: a large prime number (p) and a generator (g). These values do not need to be secret.

Step 2: Generate private keys

Alice selects a private number (a), while Bob selects a private number (b). Both keep their private values secret.

Step 3: Calculate public values

Alice calculates:

$$
A = g^a \bmod p
$$

Bob calculates:

$$
B = g^b \bmod p
$$

They exchange their public values over the network.

Step 4: Calculate the shared secret

Alice calculates:

$$
K = B^a \bmod p
$$

Bob calculates:

$$
K = A^b \bmod p
$$

Both obtain the same shared secret:

$$
K = g^{ab} \bmod p
$$

This shared secret can be processed using a key-derivation function to create suitable session keys for secure communication.

Example:

Let (p=23) and (g=5).

Alice's private key: (a=6)

Bob's private key: (b=15)

Alice's public value: (A=5^6 \bmod 23=8)

Bob's public value: (B=5^{15} \bmod 23=19)

Alice calculates (19^6 \bmod 23=2), and Bob calculates (8^{15} \bmod 23=2).

Therefore, both users obtain the same shared secret, (K=2). These small numbers are only for demonstration and are not secure for real-world communication.

B. Identify a possible security threat and suggest a solution to mitigate it.

Security Threat: Man-in-the-Middle (MitM) Attack

A Man-in-the-Middle attack occurs when an attacker secretly intercepts communication between Alice and Bob. The attacker can replace the public values exchanged during the Diffie–Hellman process.

The attacker may establish one shared secret with Alice and another with Bob. This allows the attacker to intercept, read, or modify messages passing between them. Basic Diffie–Hellman does not authenticate the identities of the communicating users.

Mitigation Techniques

Authentication: Use digital signatures or digital certificates to verify the identities of Alice and Bob.

Secure Protocols: Use an authenticated protocol such as TLS to protect the key exchange.

Strong Parameters: Use modern, approved Diffie–Hellman groups or elliptic-curve Diffie–Hellman.

Key Derivation: Process the shared secret through a secure key-derivation function before using it for encryption.

Authenticated Encryption: Use algorithms such as AES-GCM to protect messages and detect unauthorized modifications.

Private-Key Protection: Keep private values secret and use fresh private values for each session where appropriate.

Conclusion

Diffie–Hellman key exchange enables two users to establish a shared secret over an unsecured network without sending the secret directly. However, it does not independently authenticate users, making it vulnerable to Man-in-the-Middle attacks. Using digital certificates, digital signatures, secure protocols, strong parameters, and authenticated encryption helps establish secure communication. Therefore, proper authentication and key management are essential for protecting information exchanged over public networks.

References

- NIST SP 800-56A Rev. 3 – Recommendation for Pair-Wise Key-Establishment Schemes.
- RFC 8446 – The Transport Layer Security (TLS) Protocol Version 1.3.
- NIST SP 800-57 – Recommendation for Key Management.